Codebasics SQL Challenge

Q1. Provide the list of markets in which customer "Atliq Exclusive" operates its

business in the APAC region

```
SELECT market
FROM dim_customer
WHERE customer = 'Atliq Exclusive' AND region = 'APAC'
GROUP BY market
ORDER BY market;
```

market
Australia
Bangladesh
India
Indonesia
Japan
Newzealand
Philiphines
South Korea

Q2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,

unique_products_2020

unique_products_2021

percentage_chg

```
select A.unique_products_2020, B.unique_products_2021,
round((B.unique_products_2021-
A.unique_products_2020)*100/A.unique_products_2020, 2) as
percentage_chg
FROM
(
(select COUNT(DISTINCT(product_code)) as unique_products_2020
from fact_sales_monthly
where fiscal_year =2020) A, #245 products

(select COUNT(DISTINCT(product_code)) as unique_products_2021
from fact_sales_monthly
where fiscal_year =2021) B #334 products
)
```

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33

Q3 Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields: segment product_count

```
select segment, COUNT(DISTINCT(product)) as product_count
from dim_product
group by segment
order by product_count desc;
```

segment	product_count
Accessories	20
Peripherals	20
Notebook	17
Storage	9
Desktop	4
Networking	3

Q4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,

segment product_count_2020 product_count_2021 difference

```
SELECT
   dp.segment,
   COUNT(DISTINCT CASE WHEN f.fiscal_year = 2020 THEN f.product_code END)
AS product_count_2020,
   COUNT(DISTINCT CASE WHEN f.fiscal_year = 2021 THEN f.product_code END)
AS product_count_2021,
   COUNT(DISTINCT CASE WHEN f.fiscal_year = 2021 THEN f.product_code END)
   -
   COUNT(DISTINCT CASE WHEN f.fiscal_year = 2020 THEN f.product_code END)
   AS difference
FROM fact_sales_monthly f
JOIN dim_product dp
   ON f.product_code = dp.product_code
GROUP BY dp.segment
ORDER BY difference DESC;
```

segment	product_count_2020	product_count_2021	difference
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

Q5. Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields,

product_code product manufacturing_cost

```
select f.product_code, p.product, f.manufacturing_cost
from fact_manufacturing_cost f
join dim_product p
using(product_code)
where manufacturing_cost
in (
        Select max(manufacturing_cost) from fact_manufacturing_cost
#240.5364
        UNION
        select min(manufacturing_cost) from fact_manufacturing_cost #0.8920
   )
order by manufacturing_cost desc
```

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Q6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,

customer_code customer average_discount_percentage

```
select f.customer_code, c.customer,
ROUND(avg(pre_invoice_discount_pct)*100, 2) as avg_discount_percentage
from fact_pre_invoice_deductions f
join dim_customer c
using(customer_code)
where f.fiscal_year =2021
and c.market='India'
group by f.customer_code, c.customer
order by avg_discount_percentage DESC
limit 5;
```

customer_code	customer	avg_discount_percentage
90002009	Flipkart	30.83
90002006	Viveks	30.38
90002003	Ezone	30.28
90002002	Croma	30.25
90002016	Amazon	29.33

/*Q7. Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns: Month Year Gross sales Amount

fact_sales_monthly: date, product_code, customer_code, sold_quantity, fiscal_year

dim_customer: customer_code, customer, platform, channel, market, sub_zone, region

fact_gross_price: product_code, fiscal_year, gross_price

Gross sales amount =gross_price * sold_qty (fact_gross_price, fact_sales_monthly)

```
select concat(monthname(fs.date), ' (', year(fs.date), ')') as MONTH, fs.fiscal_year,

round((g.gross_price*fs.sold_quantity), 2) AS Gross_sales_Amount

from fact_sales_monthly fs

join fact_gross_price g USING (product_code)

join dim_customer c USING(customer_code)

where c.customer='Atliq Exclusive'

group by month

order by fiscal_year
```

MONTH	fiscal_year	Gross_sales_Amount
September (2019)	2020	2610.85
October (2019)	2020	762.29
November (2019)	2020	3620.89
December (2019)	2020	1372.13
January (2020)	2020	3068.23
February (2020)	2020	514.55

Q8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,

Quarter total_sold_quantity

```
SELECT
   QUARTER(date) AS Quarter,
   SUM(sold_quantity) AS total_sold_quantity
FROM fact_sales_monthly
WHERE YEAR(date) = 2020
GROUP BY Quarter
ORDER BY total_sold_quantity DESC;
```

Quarter	total_sold_quantity
4	17447125
3	7137551
1	3704398
2	3395899

Q9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel gross_sales_mln percentage

```
select c.channel,
round(sum(g.gross_price*fs.sold_quantity)/1000000, 2) AS Gross_sales_mln
/*round(
                sum(fs.sold_quantity *g.gross_price)*100/
      sum(sum(fs.sold_quantity *g.gross_price)) OVER (), 2) as percentage*/
from fact_sales_monthly fs
join fact_gross_price g USING (product_code)
join dim_customer c USING(customer_code)
where fs.fiscal_year = 2021
group by c.channel
```

channel	Gross_sales_mln
Direct	406.69
Distributor	297.18
Retailer	1924.17

/*Q.10 Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields, division product_code

```
WITH product_sales AS (
   SELECT
      dp.division,
      fs.product_code,
      SUM(fs.sold_quantity) AS total_sold_quantity
   FROM fact_sales_monthly fs
   JOIN dim_product dp
      ON fs.product_code = dp.product_code
   WHERE fs.fiscal_year = 2021
   GROUP BY dp.division, fs.product_code
)

SELECT *
FROM (
   SELECT *,
      ROW_NUMBER() OVER (
         PARTITION BY division
         ORDER BY total_sold_quantity DESC
      ) AS rank_in_division
   FROM product_sales
) ranked
WHERE rank_in_division <= 3;
```

division	product_code	total_sold_quantity	rank_in_division
N & S	A6720160103	701373	1
N & S	A6818160202	688003	2
N & S	A6819160203	676245	3
P & A	A2319150302	428498	1
P & A	A2520150501	419865	2
P & A	A2520150504	419471	3
PC	A4218110202	17434	1
PC	A4319110306	17280	2
PC	A4218110208	17275	3